

Non-Newtonian Fluid Tangible User Interface for Sonic Manipulations

Non-linear Kinesthetic Feedback with Oobleck in
Audio Production

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Abstract

Digital audio interfaces often lack nuanced inherent tactile feedback. Material properties are rarely considered when designing interfaces for control. This body of work explores the integration of Oobleck, a non-Newtonian fluid, as a boundary between the user's actions and sonic transformations, bridging the motor-auditory gap. Two testing sessions with 7 music students and producers revealed that Oobleck's solid to liquid states comply with the tactile gestures to resulting sound. Also, the users perceived the prototypes inclining more into manipulating existent sound, rather than generating it. Therefore, making this work inclining in post-production practices in the later stages of the design process.

However, challenges involved the material unpredictability and rapid drying over time, resulting in time dependent behaviours. This thesis revealed that non-Newtonian fluid's inherent resistances afford multiple gestural profiles. Thus, resulting in a highly expressive physical envelope, where the material samples affected the transformation of the sound and how the user perceive the manipulated sound.

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1 Introduction

Sound is generally considered an abstract, intangible form of perception through our auditory senses. Sound can manifest itself through acoustic and digital means of interaction; e.g. strumming a guitar string or dialling a knob of an audio interface. Playing an acoustic instrument and interacting directly with its acoustic components is a different form of interaction in comparison to its digital counterpart. When interacting with a digital instrument the sonic output is being transmitted through electric signals; it is relational and not mechanical in nature (Jensenius, 2007), losing its directedness when interacting with the sound.

Thus, the relationship between the actions and sound is often relational and weak because both are artificially created (Jensenius, 2007, p. 30); the interaction lacks nuanced feedback from the material themselves to support embodiment with digital audio artefacts. However, as previously stated, the sound is considered intangible, but it has materiality. Meelberg (2023) presented sound as a temporal phenomenon, vibration where all its manifestations are resonances that constitute materiality itself. This perspective shifted the view of sound as a material, something that can be controlled and shaped through digital means. The control, however, is predisposed to be represented through visual means.

Traditionally, composers did not have those means to sculpt and interact with the sound directly; they were bound to the musical notation to create their works (Meelberg, 2023, p. 169). In current times, sound is being presented digitally as an intangible object through visual means, influenced by how traditional composers represented their works. This was also influenced by how computers were designed to precisely control and represented data through simple slides, switches and buttons.

Additionally, sonic properties can be represented in different forms of mediums, e.g. graphical and tangible interfaces, where the visual or physical manipulations of the interface represent the resulting sound produced by these audio controllers. In a digital audio workstation, sonic states are rather represented visually rather than through tangible means.

Therefore, tangible interactions are also used in audio production, e.g. Foley processes where materials such as sand, leaves and other materials are directly deformed and recorded to be later processed in the post-production stage (Woodhall, 2011). Considering this tangent between materials, actions

and sound, I took the initiative to consider dynamic materials that could enhance the performer's action. This was a probabilistic outcome that could aid sonic manipulations, reinforcing the user's perception of the material in relation to the sound.

Within this mental framework, dynamic materials, such as non-Newtonian fluids, became an interest because of the dynamic temporal states it is inherently provided by such materials. Oobleck, in this case, became an interest because of its dual states between liquid and solid; resulted from user's executed force. This distinct inherent quality reveals an opportunity for interfacing Oobleck for sonic manipulations; transforming it into a tangible user interface, TUI. Thus, it led into a "How Might We" question:

How might we approach the inherent feedback provided by the Oobleck to reinforce users' tactile perception to manipulate audio in audio production?

Formulating this problem within this context led to exploring Oobleck's non-linearity as the boundary, the main holder between the user's action and the sonic transformations. This results in the reinforcement of their own behaviour with the intangible auditory behaviours.

Thus, this initial foundation of this research leads into one question:

In what ways can the dynamic states of Oobleck be used as TUI to support audio manipulation?

Also, considering the audio production practices and the embodiment of the inherent qualities led to two other sub questions:

How does phase-changing properties of Oobleck apply in audio production?

In what ways does the material's inherent feedback sustain or de-couple the action from sounds during interactions?

2 Background

2.1 How can we perceive a medium such as sound?

Medium itself can be interpreted as a material, but also as our representation of action directed towards the world. Merleau-Ponty (1945/2005) argued that movement always takes place against a background, there is no void between these two, the hand movements contain a reference against an object (p. 159), the medium of interaction. On the other hand, McLuhan (1995) presented the medium as a series of "contents" forming the message itself;

the way we project ourselves into the medium defines how the medium represents to us.

The medium is perceived as a pattern of stimuli inherently presented from the apparatus' contents which is being interacted with. For example, a matrix of pixels creating the pattern of legible text in comparison to a light bulb. Both use the light as medium but have different means of use. This leads to either turning a light bulb on and off or changing symbols on the screen; the medium is the same, different interactions. This applies as well when using the kinaesthetic properties of Oobleck to perceive the action, since it mirrors it inherently through action.

That's how a pattern is being perceived, however, there have been presented different perspectives on perception. Svanæs (2013) presented the concept of body schema where our body includes knowledge of the structure and specifics in relation to our body. A similar perspective has also been presented by Jensenius (2007) where our own bodily actions are coupled with object-to-object actions which later results into a sound.

Particularly if you, for example, drop a glass and it hits a tiled floor the user would expect the material to produce a sound that is associated with the gesture, or attack, done to result that produced sound. Both authors display how there is embodied knowledge present in our body. One is in relation to interactions, while the other is catered to the bodily knowledge perceived as result from the sound.

Similarly, when playing an instrument, we form these associations when experiencing a mechanical or acoustic instrument. This association is named "action-sound chain" (Jensenius, 2007), a feedback loop chain where our brain processes the muscle tension when attacking an instrument resulting in a sound (Figure 1), forming a feedback loop. This feedback results not only in acoustic qualities, but mechanical as well; producing a clear link between one's actions and the sonic qualities that are produced.

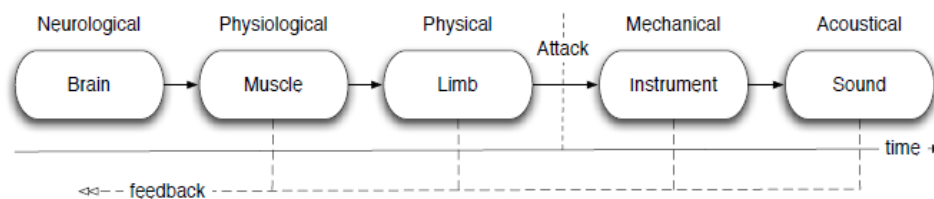


Figure 1. Action-sound feedback loop chain (Jensenius, 2007). This chain presents how performers perceive feedback from mechanical/acoustic instruments.

Thus, it creates a feedback loop that reinforces both the motor and auditory elements called *motor-mimetic cognition* (Godøy, 2003, p. 318). Motor-mimetic cognition is an inherent cognitive process where people translate musical sound into visual mental images, simulating their own actions to the responding sounds (ibid.). This reinforces user's understanding of their own

actions to the sound; essentially attributing actions to sonic producing materials.

Considering this lens, we can further investigate the meaning of sound, not as just a medium, but as a medium in relation to the material. Thus, it is essential to understand the main material investigated in this thesis.

2.2 What is Oobleck?

Oobleck, borrowed from Bartholomew and the Oobleck book by Dr. Seuss (National Institute of Standards and Technology, 2016), is a material made from a slurry of cornstarch and water, a Non-Newtonian fluid. It is also capable of changing its property to become a solid and revert to a viscous liquid once the stress produced is relieved (Ibid. , 2016) by the user.

Thus, its dual properties could potentially reinforce one's actions through its inherent affordances given by these solid/liquid states. As previously discussed, sound needs reinforcement to understand sonic manipulation, like how mechanical, material centric, acoustic artifacts already provide; through resonance.

It can be argued that Oobleck is a very complex material, but to some extent there are different approaches it can be designed for sonic manipulations. There is no extensive research on using such material as means of control, but rather a tactile apparatus to mediate with patterns, which I will later exemplify in the **Related Work**.

2.3 Approaching Oobleck as TUI

In this section I am providing how dynamic materials have been approached in the field of interaction design. This should later aid in approaching Oobleck, which is also an organic dynamic material.

One important aspect of dynamically changing interfaces is that it can be represented as either a dynamic input or an output for interactions (Coelho & Zigelbaum, 2011). Ibid. (2011, p. 163) presented a concrete breakdown of the properties of shape-changing materials, which contain:

- *Deformation Strength and Power Requirements:* This property plays a role in determining how the material is interfaced to electronic circuitry and controlled, as well as the limit in size and mobility.
- *Number of Memory shapes:* The number of active memory shapes determines how many physical configurations a material can take and if it requires a counter-actuator to return to its original shape.
- *Reversibility:* The capacity of shape changing materials to fully recover back to its shape transitions without considerable decay. This property also considers its reversibility in case the material is losing its shape changing properties.

- *Input Stimulus: The nature of stimulus required to trigger the shape change.*
- *Consistency: The material physical state (liquid/solid) which also plays a role in how you can apply such materials.*

Using this lens, Oobleck can be dissected in these appropriate properties to consider when designing such material to reinforce sonic manipulation. Also, the author mostly intended to present not only properties but possible kinetic affordances which can dynamically represent the dynamic data, in our case the digital sonic output of the TUI. Therefore, I intend to use this dissection later in the design process, to get a better understanding of possible approaches.

Also, Ishii et al. (2012) presented a different perspective, Radical Atoms, an extremely malleable and dynamic physical material that is coupled bidirectionally, representing both the digital states and its physical changes (Figure 2). Thus, the authors considered that Radical atoms should transform its shape to reflect the user input and its digital state, where it also conforms to the material and user input (p. 45).

Although Oobleck could inherently mirror its dual state to the user's input, this is not meant to aid in making a Radical atom, but rather to reference this framework for its own non-linear nature. The digital state should mirror its transformation, while the transformation informs the user; making the material the centre in the interaction.

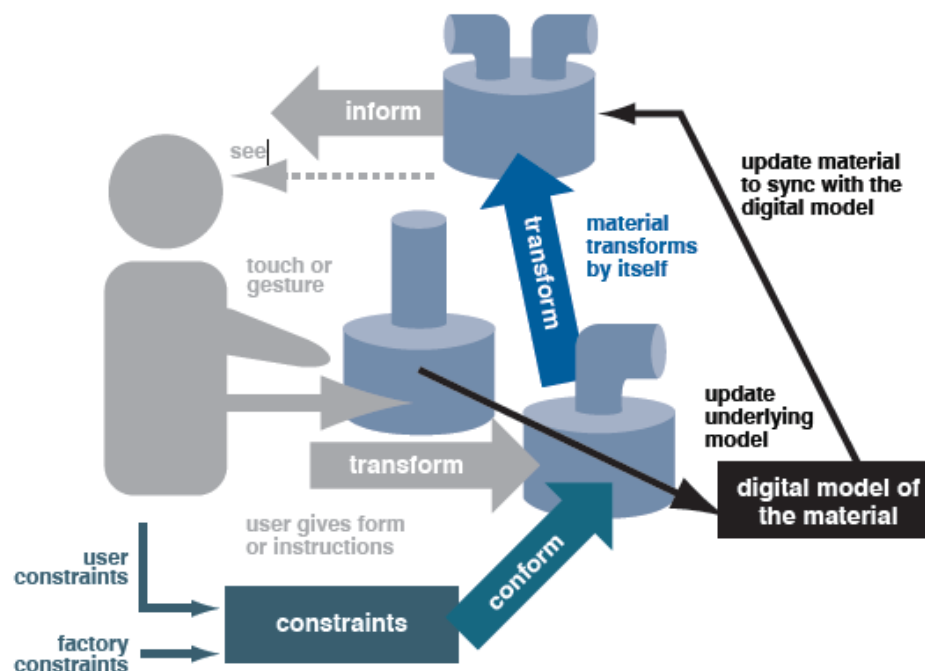


Figure 2. Radical Atom Interaction model through direct interactions with the tangible interface (Ishii et al., 2012). The user directly manipulates the artifact through touch; the artifact conforms to its constraints while also transforming both the data and it to inform the user; constantly updating its state.

Additionally, it must inform the user of its transformation capabilities through dynamic affordances through probing user's gestures with the material. Zheng et al. (2022) used a similar approach, revealing that their probing approach of the material revealed how participants engage with their material samples (p. 979). This resulting in finding the probes possible gestures and affordances. Thus, it should reveal these themes from Oobleck when testing its probes, informing for the later stages of the design process.

Ishii et al. (2012) and Coelho & Zigelbaum (2011) lenses presented the possibility of viewing Oobleck as the kinaesthetic reference from its input. Thus, the material can be the background of the interaction but also part of the foreground, reinforcing the user's actions. This can also inform the digital state of its sonic transformations through its physical state resulted of its natural inherent properties Oobleck offers.

This led to reframing and clarifying how the Oobleck can be approached as a tangible interface. The user performs an action on the material, which inherently provides its kinesthetic feedback (liquid/solid). However, it must lead in transforming the audio passively when deforming the Oobleck (Figure. 3).

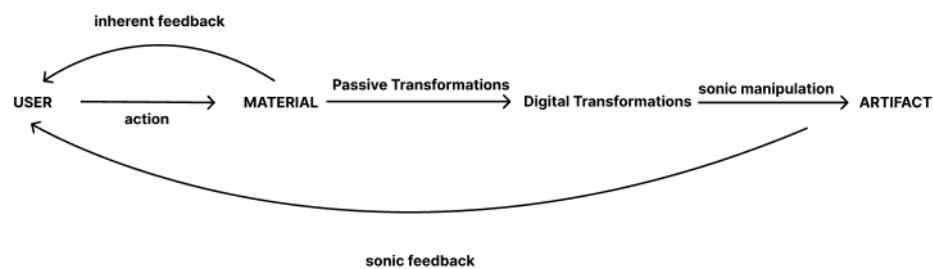


Figure 3. Inherent Passive Feedback Interaction Model. User acts with the material. Material Inherently feedback the user. The passive physical transformation is read by sensors & digital materials. Resulting in the artifact outputting sonic feedback to the user.

2.4 Computational Composites

It has been framed how users perceived their action to the sound and how Oobleck can be used as the medium for interaction. However, the digital transformations must be considered as a material part of the Oobleck. This leads to a structured interaction between the material, its digital transformation and the sonic feedback received back to the user.

Vallgård & Redström (2007) framed it as a digital composite, although it is considered immaterial, it always materializes when coupled with other tangible materials. The digital transformation embedded (e.g. sensors readings) into the material also becomes a material, although immaterial by itself, it infuses properties which make it a material composite; resulting in various effects.

The output is the expressive side of the material, while the input is the possibility of moderating that expression. The material is the constrain, while its digital counterpart expresses itself through various ways of transforming the input infused into the material. Considering this lens, the Oobleck's transformations must be considered as the digital properties that are part of the computational composite. Thus, it is necessary to explore its dynamic forms as well as its containment of different surfaces and digital sketches which builds new interactive properties.

2.5 MIDI protocol & Post-production in Audio Production

This section is meant to explain the technical terms within the domain and explain how I plan to design with this non-standard material. Also, this section introduces post-production, which later became centred and motivated from the findings in the **First Cycle** chapter.

Audio Production is currently the domain where producers digitally process and produce audio for different applications, e.g. filmmaking, advertisements, music production. The base tool audio producers use is the Digital Audio Workstation, or a DAW. This workstation is catered for producers to record, manipulate and synchronize sound for their productions (Hollin, 2026).

Also, the post-production stage is where all recorded and produced audio is finally altered (Woodhall, 2011). Audio can be used for different means of production, either being manipulated, recorded or synchronized in specific context. E.g. synchronizing and editing recorded sounds to a moving image (Woodhall, 2011, p. 231) or digitally manipulating, recording and generating sounds for music production (Hollin, 2026).

This stage has both expressivities introduced by producers, where recordings and instruments can't fully bring out, but also precise control over the final audio catered for different audio mediums (headphones, speakers, phones etc.). Later in the design process, these expressive practices are highlighted in my aforementioned chapter. This stage is another layer on the top of existing content and by adding effects to express a particular feel, for the listeners, or signature stylistic theme of the artist.

However, in post-production practices, e.g. mixing, mastering, the motor-mimetic qualities are broken; e.g. a producer moving their mouse to change the reverb's decay with no physical resistance. Considering the shear thickening properties of the Oobleck. I argue that the physical properties can be used as audio parameters in relationship to these feedback elements presented in **Table 1**.

Table 1. The Interaction-Feedback list is attributed to each actant of the interaction. It showcases each actant in relation to the interaction and its feedback. The user physically manipulated the Oobleck, the resistance is being felt directly by the user as a result from the gesture. The sensors are reading these ongoing states resulting in a dynamically transforming the audio effect.

ACTANTS	INTERACTION	FEEDBACK
USER	Physical manipulation	Kinesthetic Resistance and texture change.
OOBLECK	Transition between liquid and solid states	Passive Transformations The physical changes of the material and its tactile state.
ARTEFACT (DAW)	Sensors (e.g. Piezo, FSR, Camera tracker, Gyro) capturing the material state	Auditory Digital sonic manipulations (e.g. compression, distortion, resonance) based on viscosity.

Additionally, the most reliable tools for audio production use as basis the MIDI protocol, or Musical Instrument Digital interface. This protocol is a standard communication protocol between controllers, computers and sound modules, operating with channels and messages encoding notes, control changes and other musical parameters withing a DAW (Loy, 1985). This term will be frequently used in the design process, so I discern it as important to include.

2.6 Related Work

2.6.1 Resonance and Kinetic Tactile properties

Thompson (2018) “Ferro-Oobleck” has used non-Newtonian fluids as an interface for haptic communication. However, their focus was on actuation

and how patterns are transmitted to the user as a haptic interface (Figure 4). This research pivots away from using it as an output, but rather to explore the material's inherent kinesthetic resistance as an input for sonic manipulation. By centring on the user's tactile feedback during its deformation, I aim to address the decoupled kinetic nature of current digital audio interfaces.



Figure 4. Ferro-Oobleck Interface (Thompson, 2018). This is the interface that uses magnetic actuators to create tactile patterns.

There have been instances where some works previously presented use of tension to aid in the interaction. Vetter (2021) presented a tangible tension string prototype (Figure 5), using mechanical springs sliders to visually represent the sound envelopes and equalizer settings in the DAW. However, the author focuses on the functionality but not its kinetic properties. Focusing on displaying the data is not the point of this thesis, but to use the material's inherent nature to solicit the state of the data.

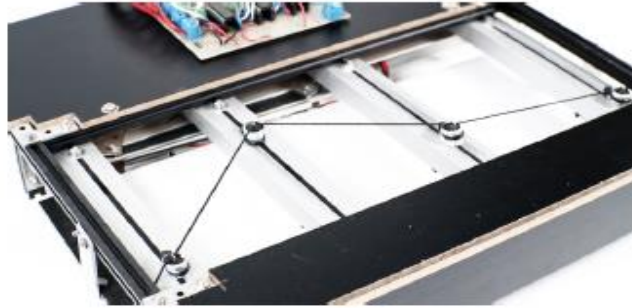


Figure 4: Tangible string prototype.

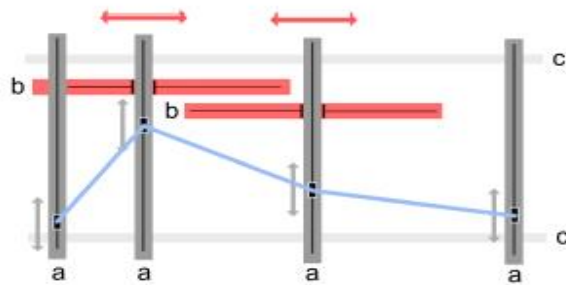


Figure 5. Tangible String Prototype (Vetter, 2021). The third prototype used as an ADSR envelope, using the strings to represent the shape of the envelope. The top apparatus has 4 slides holding the string, resulting in using the inherent tension of the string.

Similarly, Aly et al. (2018) showed that vibrational resonance of various materials can be used to trigger and select footstep sounds through their artifact (Figure 6). The acoustic resonance of the materials is the inherent feedback used as the medium of user's direct manipulation. In this rhetoric, Oobleck displays other non-acoustic properties that are much more nuanced. For example, viscosity and tactile resistance, resulting in more abstract forms of controlling and understanding the sonic transformation.

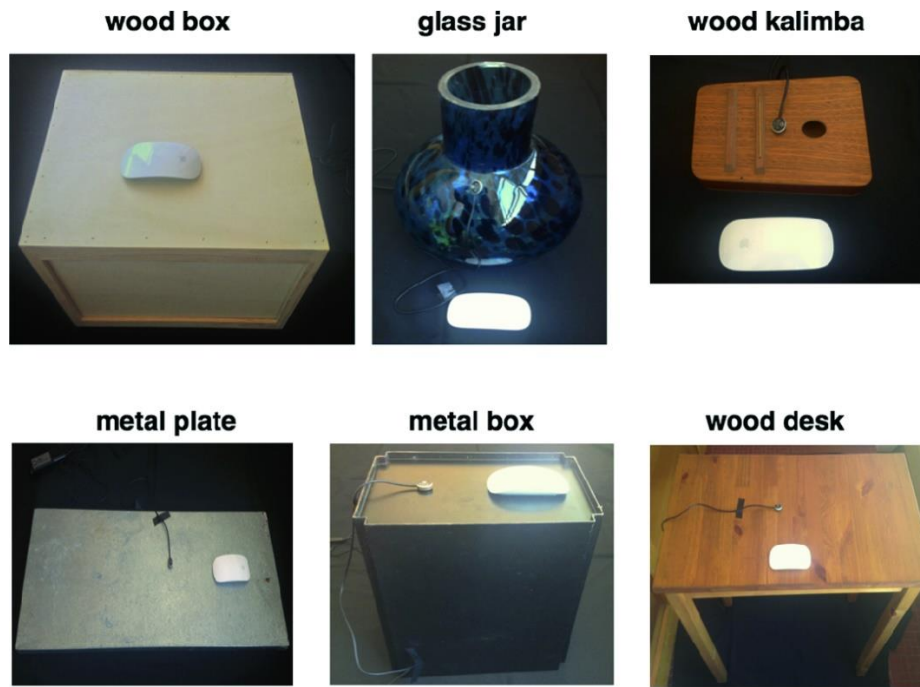


Figure 6. The *DIGIT*'s containers used to select digitally footstep audio samples using acoustic properties to sort through mapping the recorded gesture to a particular sample.

2.6.2 Expressive MIDI Artifacts

There are cases where inherent qualities have been used to feedback the user's affective input for sonic manipulations. Lamb & Robertson (2011) designed the silicone of their *Seaboard* keyboard as a linear spring, using the proportional pressure of the users applied to its silicone displacement (Figure 7). Thus, it affords linear malleability, where the user physically manipulates the sound by feeling the limits of the silicone through compression.



Figure 7. *Seaboard Keyboard's* silicon surface. (Lamb & Robertson, 2011). The linear qualities of this surface, the linear spring effect, which feedback the user of their physical manipulations.

Also, Touché¹ (Expressive E, 2017) is a touch-based interface where the inherent feedback is given by the cylindrical spring mechanism. Unlike the silicon rubber of the Seaboard, Touché mechanical tension informs the user of the digital sonic state through linear resistances. Both works are relying solely on their linear material properties providing inherent feedback.

However, Oobleck is non-linear and very dynamic. To approach such material leads to a dynamic change of resistances from liquid to solid where it solely depends on the user's direct physical interactions.

Thus, the methodological approach is also important to consider when shifting our lenses on the material non-linear physical resistances. This approach has to consider both dynamic affordances and the materiality that is central to this thesis.

2.7 Limitations

This thesis is limited to audio production, specifically post-production practices in music production and soundscapes. The nature of this body of work shifted further in those two practices, where already composed songs and soundscapes are configured and performed on. Therefore, this decision is later motivated to focus on the less accuracy and more into the expressive practices in post-production. Also, the project is more exploratory with the Oobleck with the focus on its inherent feedback at its centre. Thus, limiting this further into exploring the material's properties in relation to auditory perception of the users.

3 Methodology

The overall methodological approach in this thesis is using Research through design, this is centred to formulate hypothesis by testing prototypes built as means of inquiry, for the next design cycle (Koskinen et al. 2013). The hypothesis is produced on answering the research questions initially formulated, revealing a new hypothesis in the process every cycle. This centers on the material as the base and its manifestations as the vehicle on which we will answer our research questions related to its non-linearity and inherent feedback.

The reason I am using this methodology is because of the nature of my thesis; it's a material exploration. Such material needs to be reformulated in detail and analysed throughout each stage. This will later be explained on how these methods are structured overall.

¹ [E Touché Video](#)

3.1 Attributing and Think-aloud Protocols

Describing interactions and building a vocabulary for Oobleck is challenging. It requires a systematic approach, such as Lenz et al. (2013) interaction vocabulary. This vocabulary is a list of attributes which describe the feel of the interaction (p. 128). Although this is a tool for communicating the interactive qualities between designers, this vocabulary can be constructed for the average user as well. For example, the interaction between a hammer and a nail is “apparent”, while its material sturdiness (hammer) and the solid feel and small size (nail) can feel “powerful” and “precise”.

Thus, it leads to a possibility to not only use it to describe interaction but also feature Oobleck’s properties. This will form to the characteristics that link back to the future interaction modalities designed later in the process. However, rather than setting a language, I will allow participants to describe the interactions within their own vocabulary.

Another method that will be used for this purpose is the Think-aloud protocol. The think aloud protocol is conducted to give access to researchers and designers to what occurs in people’s minds during their thought process (Hornbæk et al., 2025, p. 742) . This will be used to validate their short-term memory of their interactions and later their descriptions of the interaction. This approach will give a clear idea of how user’s felt in the moment and how they processed the interactions with the material.

3.2 Interaction Driven Design vs. Material Driven Design

This section presents my view on the methodological approach of these two methods in context of my exploratory approach of the Oobleck; it is used for explaining the use of these methods together.

3.2.1 Interaction Driven Design

Maeng et al. (2012) presented a method of finding possibilities of designing patterns through proposing interaction as a starting point. Traditionally user-driven products have been formulated to understand the needs and its interaction, or the function is later formed within that context of use (Figure 8). Therefore, the authors proposed a structure stemming from the interaction as the standing point that led into designing the interactive prototypes (Figure 9).

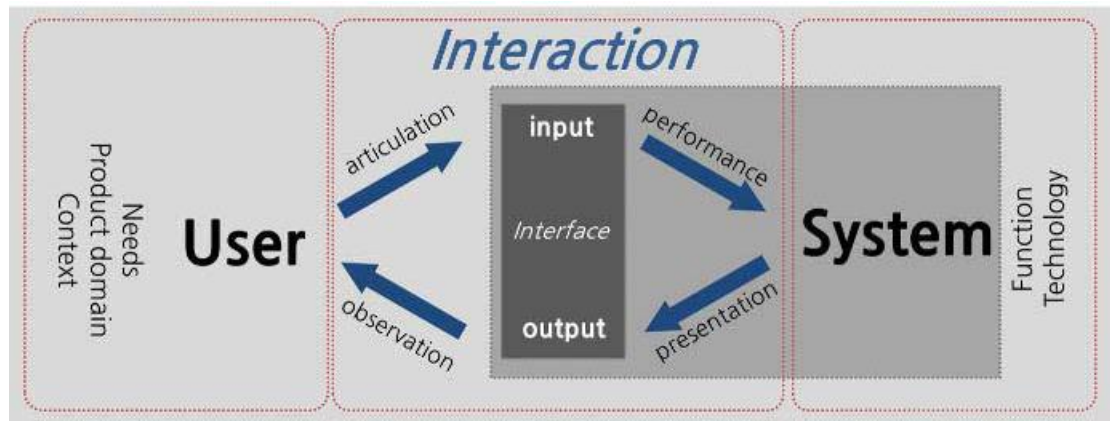


Figure 8. Structure of the interactive product. (Maeng et al., 2012)

Maeng et al. (2012) had tested this method with other designers. Consequently, the participants shaped their interactions through the boundary of a material, e.g. the rolling, spinning with the use of hinges or throwing/catching with a mobile phone to bed (p. 454). The materials afford interaction movements; their properties are the ones informing the user, visually and tangibly.

Considering that this thesis project is catered to explore Oobleck, it would be appropriate to start with a material driven approach; so that I can focus more on the interactions resulted through this process.

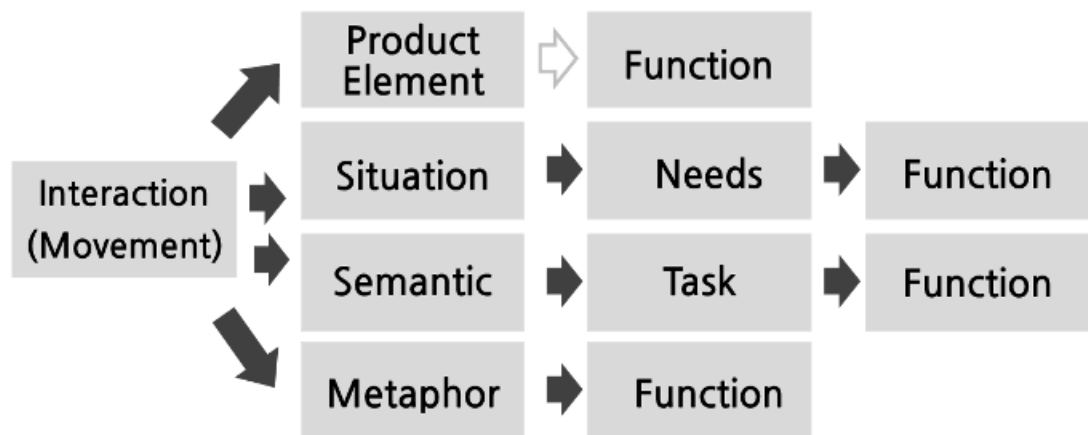


Figure 9. Ideation patterns using interactions as starting points (Maeng et al., 2012).

3.2.2 Material Driven Design

Karana et al. (2015) proposed this method that has its focus on material experience and understanding the material in the earlier stages. The author considers that designers should tinker with the material to form insights on possible material affordances, as well as technical/mechanical properties (p.41). Within this perspective, understanding the material and experimenting plays a role in forming an overall insight of how the material behaves to find possible interactions.

The overall design process is circular, and it can be seen in Figure 10. Within this thesis context I will focus on the first three processes. Understanding the material, creating the vision of material experience and manifesting material experiences.

The first stage is catered to experiment and understand the material you are working with through benchmarking and experimentation. The second stage is catered to reflect on the material's characteristics. Whereas the third stage has the role of finding manifestations within those characteristics, creating an experience pattern (Figure 11). The last process manifestation focuses on how the interaction between the user, and the material is created.

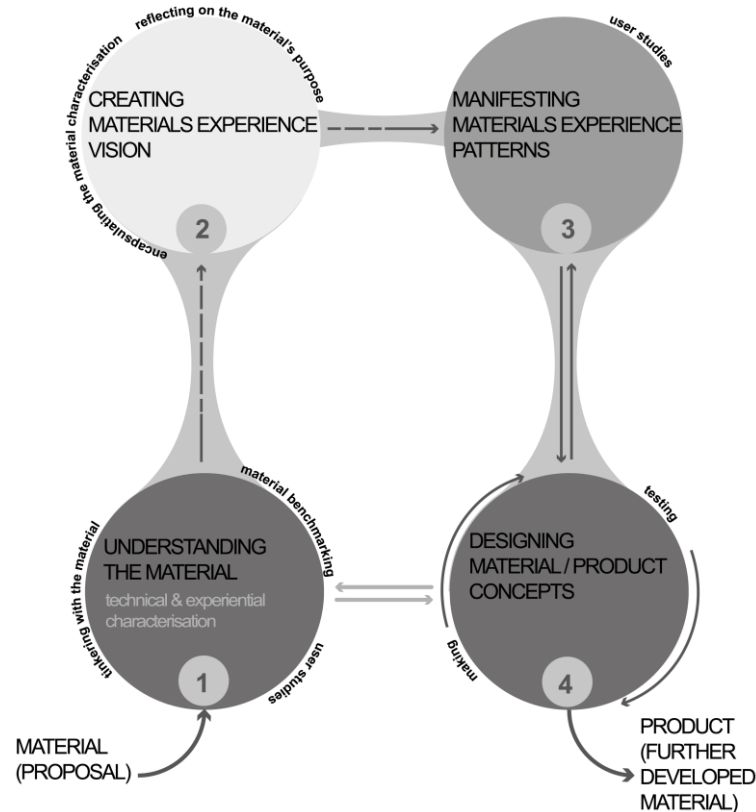


Figure 10. Material Driven Design method (Karana et al., 2015). The method is center on understanding the material through benchmarking and user studies. Later in the 2nd and 3rd are focused on reflecting and creating a vision of material experience to manifest.

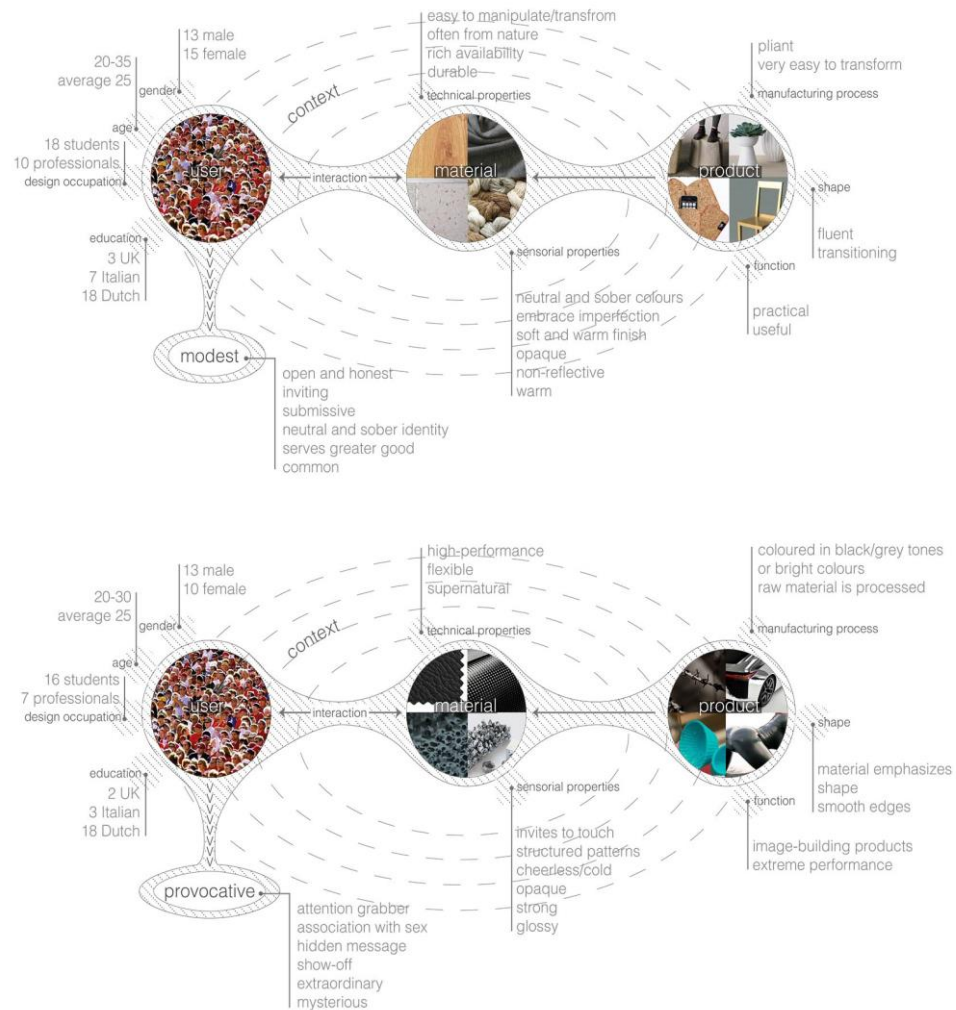


Figure 11. Material Experience Patterns. (Karana et al., 2015) . The resulting patterns that designers should conclude during the third stage of the method. (Interactions, material properties, possible products, shape, function, sensorial properties of the material etc.)

Within this context we are working with a design catered for research and not for a product. The main reason I am using this method is to find the affordances that Oobleck and its possible characteristics reveal.

Otherwise, it would lose its purpose for research through design, it would direct to a product, something not catered for this work. Thus, the methods will focus on the interaction itself and how material properties affect user's perception of the material-sound relation in the interaction. Considering that route of investigation, I considered combining Research through Design, Material Driven Design and Interaction Driven Design (Figure 12).

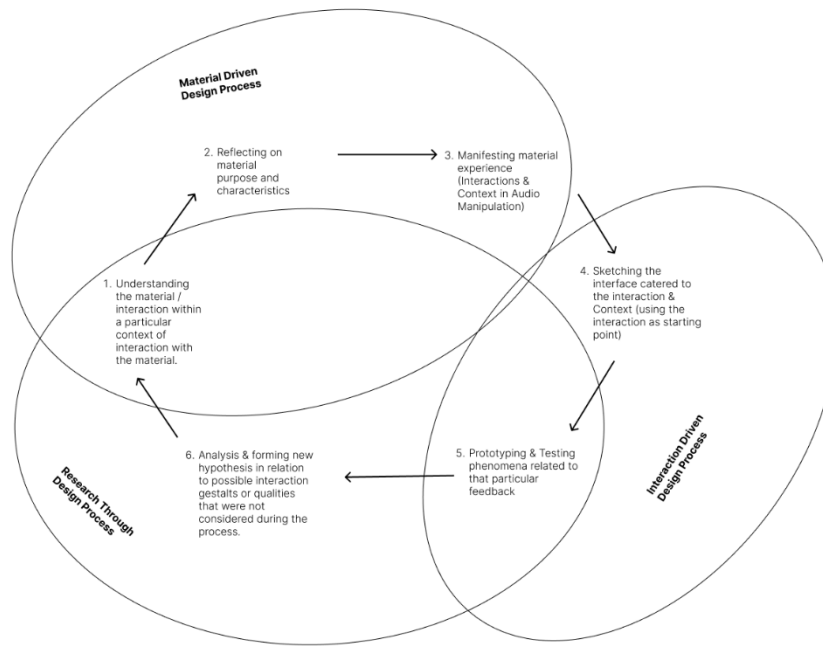


Figure 12. My hybrid methodological approach. Using elements of all three methods to consider the qualitative aspect of the interaction.

Within this perspective, the first three stages of the material driven design play the role of understanding the material. The fourth and the fifth are interaction driven design stages. These have the role to exploring the interaction formulated from the first three stages and its context of use. While the rest of the stages play the role for prototyping in means of inquiry, resulting into new hypothesis from the interactions that are reveal later in the process.

4 Design Process

This section is presenting the design process and decisions following the methodological approach I choose to take. There are some deviations in the design process, however, it will be addressed later. Since the process is cyclical, I will name each recursive process a cycle.

4.1 Ethical Considerations

The research in this thesis will follow the guidelines for ethical research ruled by Swedish Research Council and Åkerman (2024). One fundamental principle related to our thesis is to respect research participants and others involved (ibid.,2024, p. 56). Our role within this principle is to consider possibilities of discomfort, as well as regarding safety, privacy and working

conditions. Following these guidelines, I gave all participants that been part of the tests an information letter that is also form; this can be seen in **Appendix**.

Thus, one aspect that might be a problem is sanitation, since there will be quite a lot of people touching the organic material. This can also be a risk of bacterial hazards through cuts or open wounds, which made me consider change the sample every use.

4.2 First Cycle

4.2.1 Benchmarking

To assess Oobleck's temporal viscous properties, I am following Coelho & Zigelbaum (2011) frame of shape change interface to guide on possible interactions. However, benchmarking and understanding the material must centre on it's sheer-thickening properties and its conductivity.

The benchmarking its physical states was mostly done by bodystorming and interacting with the material in different size containers (Figure 13). The criteria were to reveal viscous properties, tactile qualities, it's short and long-term transformations.



Figure 13. The three containers. A large plastic container (Wide container), a medium ceramic bowl and a small shot glass.

For its conductive properties, I chose to use a capacitive sensor to read using an Arduino Nano BLE² and a capacitive touch module³ to assess how sensitive the material is to capacitive touch.

4.2.2 Results and reflection

The benchmark revealed that:

- Its viscosity changes not only on surface pressure, but on the pressures done internally.
- The concentration of water affects the density changes of the material from a slower transition to a faster one.
- It binds to the hand when sinking in the material.
- The surface becomes easier to slide with the finger on a wider surface. When the force increases, it shifts back to its powder-like texture.

However, the raw data showed that over time its capacitive properties are less sensitive, because the cornstarch was settling at the bottom of the container. It resulted in inconsistent data on conductivity over time in relation to the material. Therefore, I decided to focus on the viscosity and not continue with the other property.

Considering their aspects, the results also revealed these concrete material properties of Oobleck:

- *Deformation Strength and Power Requirements:* It requires pressure-based interactions mainly, since conductivity is much more inconsistent in that regard, thus it requires a pressure sensor. Also, the size of the container must be in smaller containers, because it tends for the water to build up quite inconsistently in bigger containers.
- *Number of Memory shapes:* It has 4 “memory” shapes. Its dual states represent the 2 memory shapes that transition from one another when applying pressure. However, when the material is not engaged over time, the cornstarch settles and the water dry leaving the cornstarch in its powered state.
- *Reversibility:* It can reverse back to its Oobleck affordances when introducing more water into the mixture.
- *Input Stimulus:* Force applied in and on the surface.
- *Consistency:* As known, Oobleck changes its physical states from liquid to solid when force is being applied.

However, it's input stimulus resulted on more nuanced results. The force done in and out of the material resulted different transition speeds. The larger

² <https://docs.arduino.cc/hardware/nano-33-ble/>

³ <https://learn.adafruit.com/adafruit-mp121-12-key-capacitive-touch-sensor-breakout-tutorial/overview>

container displayed a faster transition because of the height and volume of the sample. The medium and smaller container displayed the binding qualities as well and different transition speed because of the water content.

The small container dried the fastest in the process and led a slower transition. While the medium and the larger containers, because of the higher moisture level led to faster transitions between solid/liquid inherent transformations. Conclusively, the material transformations are dependent on the size of the container, moisture, and the force done by user's internally and externally.

4.2.3 Manifesting Material Experience & Prototyping

These inherent properties dependent on their containments became clearer, once I used force sensor resistors, or FSR underneath the samples to read pressure. Therefore, I decided to investigate Oobleck's containments and surfaces, creating material composites, varying its properties through these "membranes". Which later is attributed with the computational properties (interactivity component) later in the process, resulting in computational composite.

This resulted in experimenting on different surfaces (Figure 14), each resulting in different behaviours. The freeform is the contained Oobleck without any external surface. The blue elastic container holds a balloon filled with the same mixture, named the elastic cover. While the wider container has a plastic zip bag; the plastic cover.

The freeform, as previously stated, has the property to stick to the finger once the user internally presses. However, the other two covers filtered the sticky quality of the Oobleck. Also introduced sustaining pressure qualities, as well as force-back from the elastic surface. This was motivated to eliminate that property that I found difficult working with when coupled with the sensors.



Figure 14. The three Oobleck Composites. The Elastic cover, the plastic cover and the freeform (Left to right). The blue elastic cover resulted in most sustaining elastic properties, while also decaying in pressures over time. The one in centre resulted (plastic cover) resulted slower decay, why also denser consistencies.

For the first prototype I chose to use a force sensor resistor or FSR, an Arduino Board Nano 33 BLE and the Control Surface library⁴ to convert the pressure data into MIDI CC signals. The data has been linearly mapped between 0 and 4019 to 0 and 127; MIDI CC value range. This was done to manipulate four effects in Ableton Live 12⁵, the DAW, to directly manipulate the sonic qualities.

Also, the current configuration was made to follow the material transformations at first. Thus, assessing how the digital-material composite “perceiving” the Oobleck. Therefore, considering the digital transformation, I had to monitor⁶ the raw data for each session and material configuration, to later store and interpret the readings; mainly I used Arduino IDE Serial Plotter⁷ to read the data.

⁴ <https://github.com/tttapa/Control-Surface>

⁵ <https://www.ableton.com/en/trial/>

⁶ https://github.com/DrakeMech/ThesisAppendixContent/tree/main/Tools/FSR_Data_Recorder

⁷ <https://docs.arduino.cc/software/ide-v2/tutorials/ide-v2-serial-plotter/>

For the test I had to assess four main situations:

- Filtering with a low pass filter a gravel walking generated audio⁸.
- Configuring an oscillator level of a wooden steps synth resulting in a damper to warmer sound⁹.
- Configuring the resonance of the sample, using the Corpus filter. This results in chime to a metal horn like quality, resulted from a wind sample.¹⁰
- Panning a street background soundscape¹¹.

Each situation was catered to include from different levels of sonic apprehension from very apparent (panning) to hardily perceivable auditory configurations (corpus). This was mainly due to each effect quality difference in terms of sonic quality and behaviour which could lead to nuanced results.

Concluding on the reflection of the Oobleck and its own material composites (material, digital and audio). I decided to manifest these prototypes mainly for modulating, because the material afforded different sustained changes from one another.

For the prototype, the FSR was placed at the bottom of the container of each sample. Each material was tested on each situations presented and the participants were not allowed to look at the laptop screen. The reason was so that their foreground of the interaction is the sonic transformations with no additional visual feedback. While the background of the interaction was focused on the Oobleck's transformations.

Each participant is tasked to interact with each container, switching them after each situation given to all digital-material composites (Figure 15). In terms of sonic output, headphones were given to the user to test when panning. The rest of the situations were conducted with the use of a speaker.

⁸[Gravel Sound](#)

⁹[Wooden Steps](#)

¹⁰[Corpus Filter](#)

¹¹ [Panning Street Soundscape](#) (Recommend headphones!)



Figure 15. Testing setup. The FSR and the Arduino is on the right. While the headphones and the Bluetooth speaker on the left. This setup has been repeated each session resulting in approximately 45 minutes.

4.2.4 Participants

The participants in this test were music students and music producers. Each had different background and main instrument, as well as different audio production practices in their work (Table 2).

Additionally, during the test they've been video and audio recorded, while also recording the raw data of the FSR during each session. This led to individual testing sessions taking approximately 45 minutes. Additional notes have been taken after the session, because the participants started to associate with specific practices in audio production.

Table 2. Participant list (Anonymised), their instruments and practises in audio/music production. This information was given by the participants during an informal interview done before the tests.

PARTICIPANTS	INSTRUMENTS	PRACTICES
P1 (Performer)	Classic Guitar	Recording, Mix and master.
P2 (Music Producer)	Piano, Bass, MIDI Keyboard, Acoustic Guitar etc.	Mix and master, recording, digital Foley, audio synthesizing etc.

P3 (Performer)	Violin	No experience in any audio production.
P4 (Performer)	Saxophone	Recording and synchronizing audio.
P5 (Music Producer)	MIDI keyboard, MIDI launch pads and controller	Mix and master, recording, audio synthesizing etc.
P6 (Performer)	Guitar	Has just basic understanding of recording and using live tools (Amplifiers etc).

During the test, each participant is individually engaging in a discussion around the material. Therefore, they were asked to write one term describing a distinct quality of each set. Also, they've been tasked to verbalize and describe what they perceive during the interaction.

4.2.5 Results

As previously stated, I am looking not only on the dynamic states of the Oobleck to support sonic manipulations, but also how the users perceive the audio using this interface. The raw data showcased here are all resulted gestured given by all 6 participants clustered into one graph to show the patterns given by each material-sensor reading. The raw data of each session and documented notes after the tests can be seen in the **Appendix**.

The materials presented very different behaviours. The freeform presented a more immediate but unstable temporal behaviour. It revealed the most frequent interaction gestalts in the motor-mimetic loop. Thus, resulting in almost an on/off responses by the artefact sounds when users pressed harder (Figure 16). However, low resistance gestures revealed that it would not only remain in its liquid state, but also not result in any sonic changes; VIDEO¹².

Also, gestures like spinning, scrapping and partial resistances done inside the material, afforded for very apparent sonic changes, being able to sustain the changes for longer; VIDEO¹³ (Figure 17).

¹² <https://youtu.be/p8GQ7gLXrog>

¹³ <https://youtu.be/A3P4V1bKtaY>

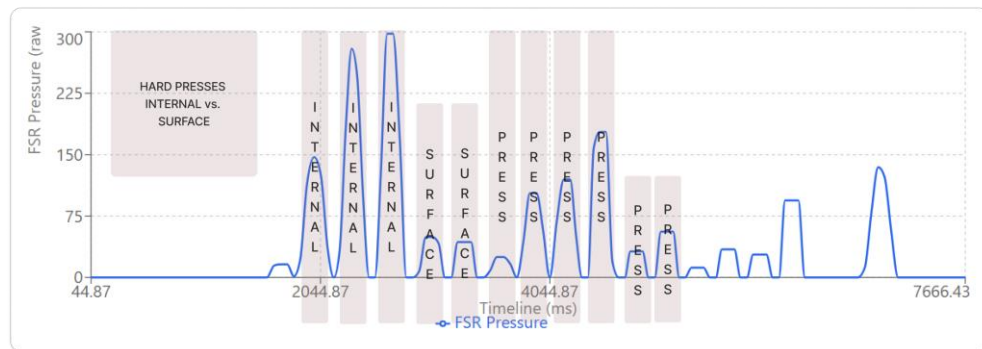


Figure 16. On/off like readings based on user's pressure. The internal presses resulted to higher peak responses in comparison to the surface-level pressures. (Higher the amplitude = higher the forces received by the sample = more apparent changes to the solid state).

Thus, the free form's inherent qualities, such as its stickiness and liquidity on both high and low energy gestures overlapped to one another. P1 and P5 prompted that there is no consistency, it “glitches” and “blips” when configuring the panning and the resonance of the sample. Similarly, P6 and P3 resulted in very gentle gestures, thus resulting to almost no sonic changes; describing the material as “sticky” and “liquid”. P4 prompted that it reminds them of working the existing sound like a clay.

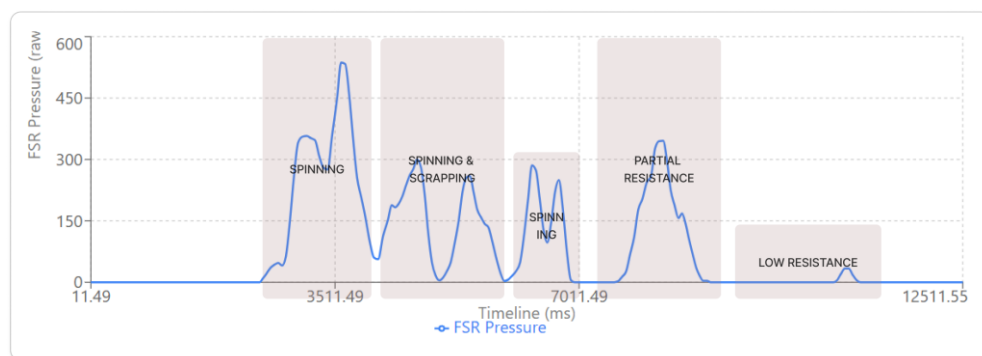


Figure 17. Spinning and Scrapping resulted in more sustainable changes. However, on lower forces given by the user, there are no apparent responses. (Higher the amplitude = higher the forces received by the sample = more apparent changes to the solid state).

The elastic setup was more continuous in its behaviour, resulting in smooth transitions between high-low force presses. Internal presses inside the balloon resulted on more sustained sonic changes (Figure 18); VIDEO¹⁴. On the other hand, surface-level interactions resulted in very short low responses (Figure 19), but also high force presses resulting a shift between gradual modulations to rapid sonic changes; VIDEO¹⁵.

¹⁴ <https://youtu.be/bQTSdLhv4SQ>

¹⁵ <https://youtu.be/Kn4nnWXkjNI>

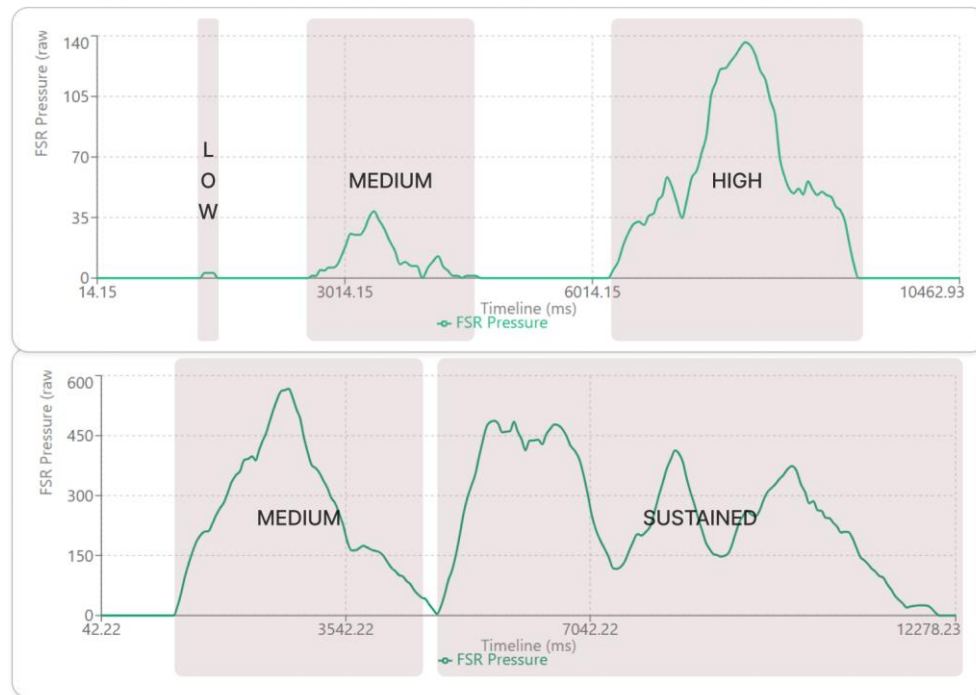


Figure 18. Internal presses inside the elastic cover resulted different degrees of sustained responses by the sensors. (Higher the amplitude = higher the forces received by the sample = more apparent changes to the solid state).

P1, P2 and P3 attributed the elastic setup as “repelling”, “controlled” and “consistent in its inconsistency”, each representing highlighted qualities they presented the setup as. The P1 and P2 associated with the elastic quality of the setup because of the repelling surface and its reversible form. P5 and P2 both did associate the elastic qualities with a “guitar pedal” and a “pitch wheel” of a keyboard because of the force back of the elastic surface. Also, P5, P3 and P2 had very similar comments on the reactivity of the material and sound over their movements.

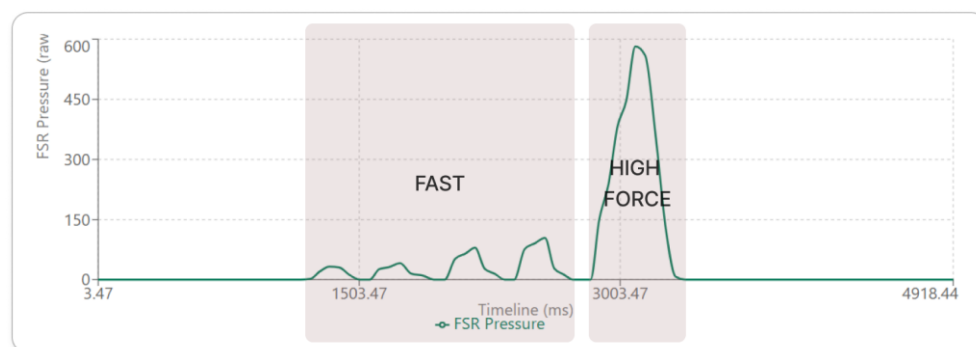


Figure 19. Surface-level presses resulted in very short responses. The sensors also result different amplitudes. (Higher the amplitude = higher the forces received by the sample = more apparent changes to the solid state).

The plastic setup on the other hand presented a delayed almost conditional behaviour. In this setup all users pressed back and forth onto the material to result on stronger sonic changes; VIDEO¹⁶.

However, there were both short-like responses and multi-finger interactions were more affordable to act, because of its shape and size (Figure 20). The transition speed of the plastic setup was also affected by the water content. P6 gentle force and the liquid form were more present and attributed as soft, whereas P2 described it as stiff and hard to bring it back to shape.



Figure 20. Plastic cover affords both short responses moving material (left-right) and sustaining interactions. (Higher the amplitude = higher the forces received by the sample = more apparent changes to the solid state).

Also, the think-loud protocol and the interaction attribute revealed different descriptors for each material composite (Table 3).

Table 3. Interaction attribute vocabulary revealed by the participants during the tests and during the discussion. This is later used in the second cycle to create metaphors for the digital sketches.

PARTICIPANTS	FREEFORM	ELASTIC	PLASTIC
P1	Gripping	Repelling; has powder texture at the bottom.	in-between
P2	Strong	Controlled	Stiff

¹⁶ <https://youtu.be/jFOf8LNGjqk>

P3	Had most control	Consistent in its inconsistency	Difficult, either super hard or soft
P4	Clay, sandy	Soft, yet tensioned	Builds up and softens
P5	Sticky	Rubbery	Flows on the side. Slow
P6	Sticky and powdery	Soft and elastic	Wet and hard to press

4.2.6 Discussion

The Oobleck's properties have been behaving differently from a surface containment and another. It presented a dynamic pattern of multiplicity of qualities associated within one material. Participants had different attributions from a material composite to another, thus resulting in different descriptors.

As previously shown, each cover revealed different associations from the materials. For example, the freeform was inconsistent for most of the users that were gentle, resulting a hidden affordance. While the elastic cover revealed rapid and gradual responses. For the plastic cover its association were more temporal to its transformations and build up. Thus, most participants described the freeform to it's liquid like state because of their gestures.

However, there have been cases where the freeform and the plastic cover where participants had different descriptors. P1 described the freeform setup as "gripping", while P2 attributed it as "strong". Also, P4 associated directly the soft feel and strong solid changes directly to the sound.

Considering these themes, Oobleck is conditioned by the user's direct manipulations. Although the material is the boundary between the user and the digital transformations, I argue that the Oobleck has its own agentic properties in the interaction. It dynamically constrains specific gestures that the user's intents to use to manipulate audio parameters. The sonic responses always varied because of the temporal behaviour resulted by material setup, thus allowing the users to get specific results (Figure 21).

Therefore, through analysing the FSRs readings I found that three factors that can be derived into nuanced interactions for the second cycle. The time it takes for the user to hold pressure into the Oobleck, the pace or rhythm in between attacks and the amplitude of the force given by the user. Thus, it gave me other possible factors to consider when sketching the interactions for this material.

Table 4. The material temporal behaviour, the overall descriptor of the user's perception of the sound and the condition of the motor-mimetic loop. Each have been documented and used at the later stages to distinguish possible metaphors in the second cycle.

Material Setup	Temporal Behavior	Sonic Perception	Motormimetic Loop
Freeform	Immediate but unstable	Apparent but chaotic	Broken
Elastic	Continuous	Smooth & learnable	Strong
Plastic	Delayed / accumulative	Emergent	Conditional

The freeform was mostly broken because of its unfiltered material nature that resulted in participants fingers stick to the container. The elastic cover, on the other hand, resulted in a strong motor mimetic loop because of its continuous transformation. The plastic cover was conditional to afford stronger sonic transformations; the participants were always conditioned to press on the sides to build up.

Although, the mapping was linear throughout the tests, each setup resulted in different qualities, showcasing different motor-mimetic loops. Therefore, there are multiple possible variants and metaphors which can be derived and embedded on top of these material composites. Thus, it made me consider cycling back into benchmarking each composite and prototype digital sketches on top of the samples.

Additionally, the gesture of the users differed in force, thus the artifacts linear mapping was a problem. Most participants expected more apparent responses. This led into creating an average mapping over time of the user to calibrate for the second cycle.

However, some deviations became apparent between participants. After testing the composites, most participants derived possible uses for this material in post-production practices. P5, P1, P2 and P4 presented the pre-recorded audio as a malleable expressive “layer”, “effect”, “clay” that is constantly changing. Thus, influenced me to reframe my research centred for post-production practices previously mentioned in the **Background**.

The agentic property of the material preserves each gesture as different interaction modality for post-production, thus resulting in a new hypothesis:

How can we use the interaction modalities given by each material setup? What is the potential uses in post-production? How would the participants approach it in their practices?

4.3 Second Cycle

Within this cycle I decided to be more domain oriented in the process, since in the previous cycle it deviated more into the material properties and control. Therefore, I decided in this cycle to test the participants on their own production files. Thus, the entire process in cantered on the domain tools, the possible applications and practices these artifacts can possibly be used for.

4.3.1 Benchmarking

Because of the importance of the mixing ratios, I had to be more concrete with the measurements of cornstarch/water ratios to have more accurate results. While the transitions speed of the Oobleck's states presented an issue, I decided to use this quality as a design advantage to control the transition speeds from a sample to another. Like in the previous cycle, I am assessing the speed of these transitions as well as the behaviour of each material container.

Thus, I decided to use a food scale to measure the percentage of water for 100g weight of Oobleck. There were made 5 samples each with different quantities of water, where I assess when the material is losing its dual properties.

4.3.2 Benchmarking Results

The variations between the solid state (the cornstarch) and the liquid state resulted in ~11% water content differences. Also, rather than having four physical states, this time it resulted in five (Table 5).

Table 5. The Oobleck's states and its ratios, as well as its water percentages resulted from the benchmark. Used to later experiment with the consistency of the inner matter of Oobleck in the composites previously made.

STATE	RATIO (cornstarch/water) (grams)	WATER %
Solid (Powder-like)	67/23	~25
Slow	73/27	~27
Standard	70/30	~30
Fast	67/33	~33
Liquid	70/35	~36

Considering these results, the Oobleck should not be lower than 27% moisture levels. Also, the slow and fast transition presented qualities that could work for time-based practices in post-productions, e.g. filters, transitions.

When exploring these samples on each cover, each presented different inherent qualities with each ratio (Table 6).

Table 6. Ratio's inherent qualities resulted from interacting with each sample. These were my descriptors resulted from bodystorming and working with the material when applying downward force.

COVER	SLOW	STANDARD	FAST
Freeform	Powder-like; hard to press; affords scratching.	In-between; affords internal pressure; stiffens when force is applied; sticky	Thick liquid; doesn't afford much pressure.
Elastic	Hard to deform; It reverses back slower.	Malleable; easy to deform; stiffens when force is applied.	Rubber-like; easy to deform; has higher force-back.
Plastic	Very slow; Hard to press.	Malleable; deforms slow; stiffens when force is applied.	Soft; easy to deform; little to no resistance.

From these qualities, it revealed each ratio has different affordances. Therefore, each inherent affordances can be used to attribute metaphors when designing these modalities of interaction.

4.3.3 Manifesting Material Experience & Prototyping

Taking this into account, I decided to use these ratios for different interactions modalities and metaphors. Also, the gestural affordances revealed that both its digital and material elements play a role in compiling these metaphors. Thus, for each cover I've included metaphors for the interaction modalities for sonic manipulations (Figure 21).

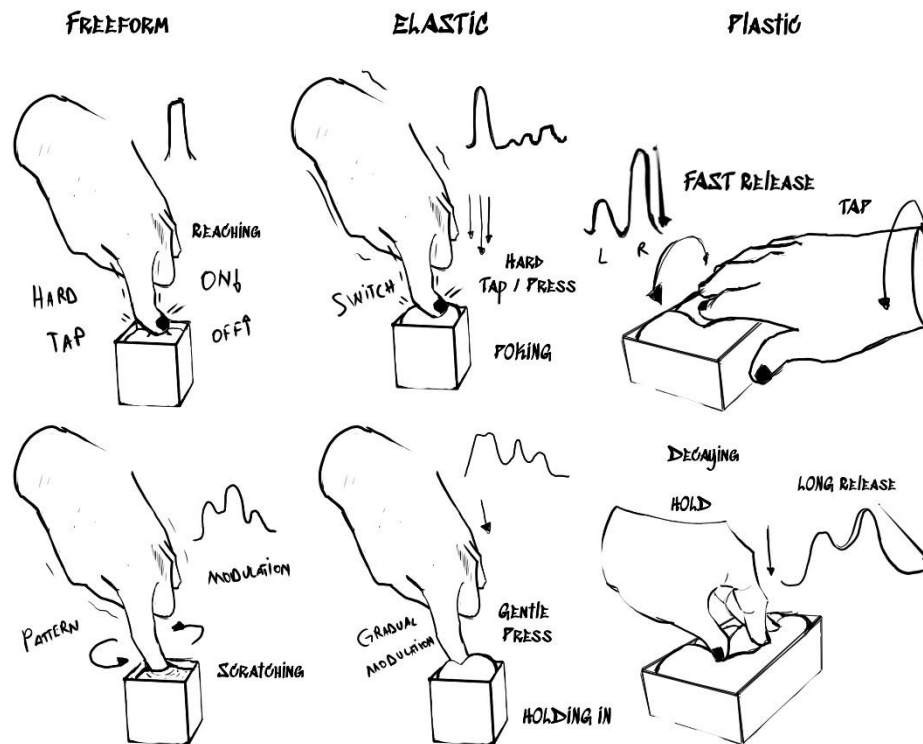


Figure 21. The sketches representing each interaction and metaphors. The one on the left is the behaviour of the “reaching”/”scrapping” metaphor. The one on the centre is the “poking”/”holding in” metaphor. The one on the right is the “decaying” holding metaphor.

The first metaphors used for the interaction was “poking” or “holding in” the elastic setup. These metaphors were appropriate because of dual switch between solid and elastic/soft affordances when manipulating the Oobleck. Thus, the I catered these properties for switching parameters in the DAW. Therefore, the solid state was attributed to signal when the user switches two audio properties. The elastic/soft state was used for modulating and sustaining the sonic changes.

The second sets of metaphors were “reaching” and “scrapping” the freeform setup. This set was much more focused on pushing in and reaching for a range between slight forces (low) and hard forces (high). Also, the stiffening is more of a mirror resulted from the movement. This will associate how far or close the user is to the second state.

However, the third metaphor for the plastic setup was much more different, “decaying”. The artifact’s material played a role through its tactile qualities to feedback the amount of decay, also the duration of the presses. Since, it affords delayed responses, I decided to use this behaviour mirroring with the decay of the sonic changes at the release of the attack from the material.

Also, these metaphors were explored and inspired from the responses given from the user’s attribute vocabulary; resulting in translating the composite

qualities to (Figure 22). The repelling and solid quality of the elastic cover was mostly associated with pedals, silicon drum pads and buttons. The sticky and claylike quality was mostly attributed to a slide that at maximum threshold it binds and gets heavy on forceful movements. The heavy and slow decaying properties were associated with a heavy knob that reverses back to its initial state.

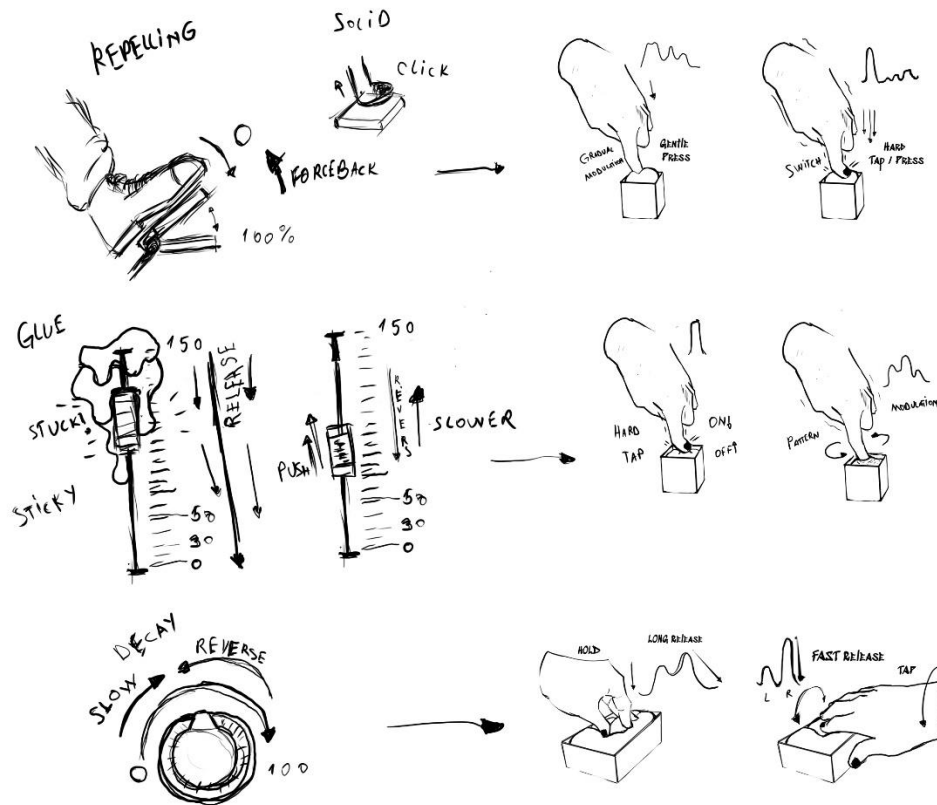


Figure 22. The qualities and behaviours inspired by already existing tangible interfaces. Each either has multiple linear properties or have qualities following the existing effects given by the Oobleck representing a particular sonic state.

4.3.4 The Casing Setup

In this stage, I had to be more careful with the risks of shortcircuiting components. Thus, this iteration has a case that hold the FSR sensors underneath the 3D printed FLEX casing (Figure 24). Also, it is used to keep the cables covered as well as the electrical components from the Oobleck.

The case and its plastic containers were inspired from Gridfinity¹⁷, since it's shape and form were compact and easy to stack. Also, the prototype has a thin elastic printed layer between the sensors and the containers to protect the sensors from the Oobleck.

¹⁷ <https://gridfinitygenerator.com/en>

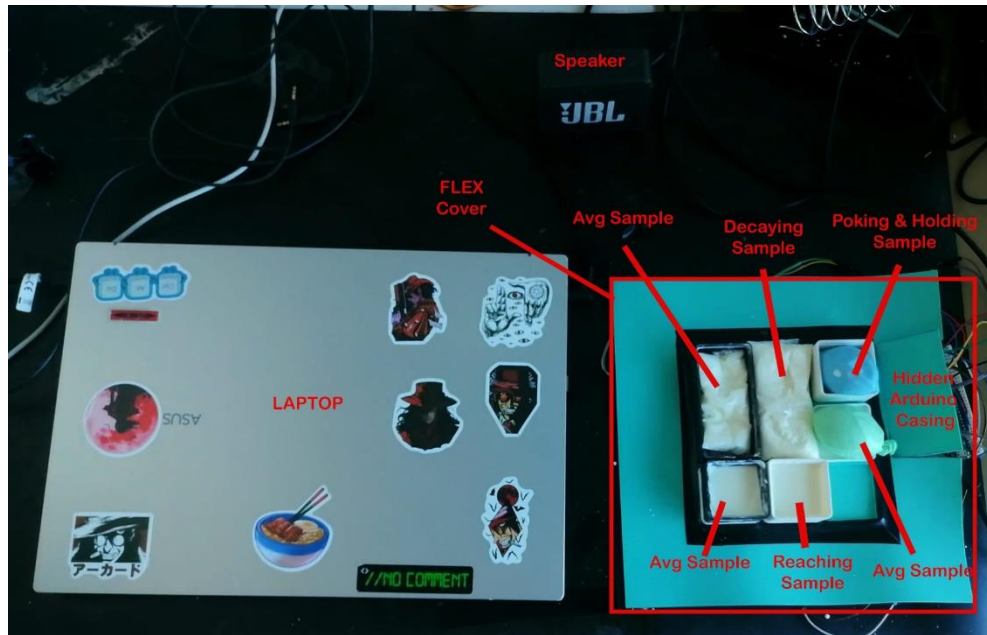


Figure 23. The test setup. On the right is the 3D Flex casing (6 inputs). Both the average sample from the previous test and the metaphor computational composites.

The prototype has 6 inputs for each material and each varied between slow and fast Oobleck samples, to match the appropriate metaphor (Table 7).

Table 7. Table showing the Oobleck's ratios for each mode and average linear mapping. These are the ratios that have been used for the linear average mappings of 3 inputs, while the other 3 inputs are the digital sketches previously mentioned.

COVER	AVERAGE MAPPING	MODES
ELASTIC	Fast	Standard
FREEFORM	Standard	Slow
PLASTIC	Fast	Slow

The “switch” mode of the elastic cover was the most appropriate for the standard Oobleck, its dual properties are already existent for that metaphor. The plastic cover, although already slow, it can still use the slowest composition for the material to decay slower; ranging from a shorter to a longer release. Additionally, the free form’s “reaching” mode is also be used from it’s on/off qualitative range; by using the slower powder-like ratio as the indicator of maximum threshold.

4.3.5 Participants

In this iteration, the participants were selected based on the level of experience. Within the previous group, only P3 and P4 didn’t have experience working in a DAW, thus I decided to bring a new participant (Table 8).

Table 8. The newly introduced participants for user testing.

PARTICIPANTS	INSTRUMENTS	PRACTICES
P7 (Audio Producer, Musician)	Piano, contrabass, drums, MIDI Keyboard etc.	Recording, Mix and master.

For this test it was re-include past participants that had experience working with post-production audio tools, specifically mixing and mastering audio. All participants were tasked to interact with the controller for approximately 20 minutes each. Also, the participants were video, and audio recorded during the testing sessions.

Additionally, the participants were allowed this time to have access to their laptop screens, while also working with their own project files. Therefore, this test was mostly directed on the post-production practice within audio production domain.

4.3.6 Results

The results have shown that participants overall used each setup for a wide range of uses:

- Freeform
 - Average mapping: P7 used it to dial between wet and dry effects, since the texture was more appropriate for that effect. Other participants used it for to filter audio
 - Peak: This material was used for manipulating track's volume and resonance on audio filter effects.
- Elastic
 - Average mapping: Modulating macros, filters, LFO's and vibrato on their pre-made synth tracks and audio samples.
 - Switch: Switching between effects and modulating the same way the other set was used for. Overly used for transitions for their compositions, while using time-based effects such as delay and reverb.
- Plastic
 - Average mapping: Used for audio amplitudes and filters; extremely used for holding or resurfacing another track; layering between audio samples.
 - Decay: Mainly used for delayed transition. Didn't had much impact in terms of consistency of use, was quite similarly used like its counterpart.

However, each displayed more interest on the elastic and plastic setup, leading to variety of sonic responses and patterns (Table 9).

Table 9. Sonic responses and pattern for each composite. It excludes the freeform because participants weren't sure for what to use it for. P7 requested not include recordings because they tested the prototype on their unreleased song. The table was derived and sourced from analysing video recording and reading my additional notes.

PARTICIPANT	ELASTIC	PLASTIC
P1 (VIDEO ¹⁸)	Changed audio levels via an LFO; Switched with a Low-Pass Filter.	Changed LFO amplitude to alter the track audio levels.
P2 (VIDEO ¹⁹)	Used a comb filter over the entire track to highlight a specific sonic effect.	Used a macro setup over some synth effects; Transitioned and layer the audio effects.
P4 (VIDEO ²⁰)	Altered the vibrato effect to a melody sample; Controlled the synthesized instrument to create a nuanced sonic output.	Used a generic filter within their project setup; Gave no further clarification due to the participant being focused on the material.
P5 (VIDEO ²¹)	Manipulated the frequency range of a low-pass filter and individual audio levels; Switched and layers both effects to transition.	Manipulated track levels for delayed transitions.
P7	Altered track feel using LFO and resonance; Switched between LFO and Low-Pass filter resonance. (Requested to not include recordings)	Manipulated the gain of the track and filter's range. (Requested to not include recordings)

The freeform was mainly used for its low/high properties to select the levels but also modulate from the other variant. However, most didn't manage to adapt that much, since their gentle force kept introducing interaction

¹⁸ <https://youtu.be/ZKTutWYOE98>

¹⁹ https://youtu.be/VUX16fV_5jg

²⁰ <https://youtu.be/P4x27KwBqME>

²¹ <https://youtu.be/hgnUIv82ruo>

gestalts. Thus, leading to them using the material rarely or even ignoring it throughout the test.

There have been situations where both consistency around each material influenced their preferences of use (Table 10).

Table 10. Situations and frequency of use between each composite. Informing for how long it has been used from each session and what effects have been used. (This was approximated through annotating the video recordings).

Skill Level	Frequency of Use (Effects)		
	(Low:<10min, Medium:<22min, High:>22min)		
Participants	Freeform Setup	Elastic Setup	Plastic Setup
P1 (Beginner)	Low (Resonance)	High (Mastering, Low-Pass Filter, LFO over Volume)	Medium (LFO & Audio Levels)
P2 (Advanced)	Low (Audio levels)	High (Comb Filter & Macros)	Medium (Low- Pass Filter, track layering and transitions)
P4(Beginner)	Low (Track Volume)	High (Vibrato)	Medium (Filter)
P5 (Advanced)	Medium (Track Volume)	High (Transitions and Low-pass filters)	Medium (Layering Audio)
P7 (Advanced)	Medium (Reverb) Wet/Dry effect)	High (LFOs & Resonance)	Medium (Gain & Filter)

After the test, I had a free discussion with each participant on where they could see these setups to be used for. P2, P7 saw this setup catered more for brainstorming effect patterns. All participants agreed that the setup is more catered for DJs, mixing & mastering and live performances, but also a tool to add expressiveness for their pre-recorded audio.

Also, P7, P5 and P2 preferred the elastic setup the most because of its sensitivity and the feel it inherently gives. P7 highlighted “the material, makes sense” and “much more tactile to a button ... it has that pillowy range”. While the plastic cover, for most, it was considered something that is slower, that you work or follow through. P7 and P2 saw it as something that “you play with them” or “you would use for transition between tracks” because of its slow pace.

4.3.7 Discussion

A common pattern given by the participants was that they got confused of what specific sonic qualities they manipulate in the elastic “poking” setup. Most participants required to preview which setting they were modulating. P5 and P7 had the easiest time to adjust to it, because of the apparent changes of the sounds they were manipulating. Thus, the theme does not apply to all cases, because each audio effect will result different levels of perceptibility.

Another pattern displayed by the participants were the use of each material on specific audio effects between one another. The plastic setup was used for smoother transition in their projects, such as filters or change of volume. This aligned to the gradual modulation afforded by the material-digital composite.

However, the freeform setup that used the *reaching* resulted in users having difficulty reaching the maximum because of the sensitivity of the sensor. This shows how the consistency should have been probed with the participants to match their strength. Thus, it would be necessary to constantly probe and reverse back to the catered state, with the risk of interrupting the participant in the process.

There have been occurrences during the tests were sustaining the transitions between slow and fast became extremely difficult. Thus, I recommend when testing these materials to focus on each individual setup; like it was done in the first cycle.

However, as previously highlighted in the results, each interaction modality linked to gesture are highly affected by qualities that majority of user preferred to not interact with e.g. Freeform’s sticky quality. Thus, these qualities will make some of the participants less likely using that setup.

Considering the transitions speeds and its ratios the slower variants not only allowed for higher sensitivity to the audio but also influenced majority of participants to use audio effects catered for configuration, e.g. track volume, reverb. The standard and faster ratios were used for effects catered for live modulation, e.g. filters, vibrato, gain. This does not include how the cover’s affect their decision on using these audio effects.

Also, users consistently used the prototype throughout the test as a brainstorming tool. P2 and P5 throughout the test look for their patterns to later record the live configurations. This aspect was unexpected, since I viewed this more of an interactive tool for live configurations, mirroring changes back to the user through its feedback. Thus, this brainstorming practice highlights Svanæs (2013) kinaesthetic creativity, where the body schema aids in generating possible futures for sonic patterns. This resulting on finding new patterns for mixing and mastering tracks in both live and in production.

For the post-production practices these practices became increasingly apparent throughout each session done with the participants. Most effects

were used on top of their works to enhance specific “feels” in their works. P7 presented the freeform as a wet/dry material directly translated on the reverb effect, resulting in duller to a dryer space. P4 and P5 used the elastic qualities of the elastic composite as a transient tool, presenting the effects over time rather than instant; like how P2 has been using it to highlight a specific sound in their tracks.

5 General Discussion

This section is used to critically analyse and bring critiques for the current and future work. This is reinstating and addressing the main research questions.

5.1 Critique on the Design Process

Also, initially considering this material as only a post-production, it led into potentially being a live performance tool, a brainstorming tool and an expressive instrument to record on top of existing content. Each participant presented these practices consistently throughout all the tests.

The interactive modalities in the other hand were underdeveloped. The freeform, although hard to work with, potentially could have used the binding quality inversely with the pressure sensor. Thus, the most gestures available have not been fully explored. This entire research revealed only interactions bounded by pressure sensors. However, a future study could also reveal possible dynamic interactions bounded by conductivity, using capacitive sensors.

Therefore, the dynamic states of Oobleck can be used to support sonic manipulations, but that is bounded by the inherent qualities, e.g. binding, slow transitional, powder-like consistency qualities. This material is always transitional from the cornstarch powder to the liquid form, affecting the material properties over time.

In terms of motor-mimetic cognition, the interaction between the action-material-sound varies because it is temporal within its context; e.g. using low-pass filter, vibrato over a sustaining gestural afforded setup can vary in attack times. However, that it must be supported by another material to filter the unwanted qualities from the Oobleck, e.g. the binding nature of the Freeform.

For the post-production, the situations become more limited to each setup. In the second cycle, the freeform presented the most problematic qualities. The binding made the user’s fingers having difficulty releasing. Therefore, the inherent feedback of this setup constantly de-coupled, resulting in constant

interaction gestalts from the users. This led to majority of users interact less with this setup, which led to less qualitative data being gathered.

Another direction could also be directed to the situations in audio production. Expanding on a larger study with professional foley artists, sound designers and mixing engineers could also yield more qualitative data. Considering how these experts can be observed adapting their bodies to Oobleck. This could also result in more insights for more specific professional settings in the industry, e.g. in post-production in film making or advertisement.

5.2 The Temporal Decay and Ephemeral nature of Composites

A critical phenomenon has been revealed during benchmarking and testing cycles. The stability of Oobleck as an interface medium changed over time. Traditional TUIs and computational composites assume a degree of material consistency; a silicone sheet or spring maintains its behaviour for longer periods of time. Oobleck, however, is ever-changing by nature, it's very time sensitive resulting to changing behaviours.

Thus, when designing a standard MIDI controller, a specific physical gesture is expected to map identically to a digital value regardless of whether the action is done at the beginning or the end of that session. With Oobleck, the evaporation of water and its precipitation of the starch led for a different threshold given by the input stimulus from the user; constantly shifting in the process. What initially was required, e.g. slow deformation maintaining its liquid state, might require an entirely different physical approach to the sample when drying.

This shifts the user experience from an already determined form of control to an adaptive, almost performative dialogue with the organic material. For post-production workflows require a longer time to work on the project, which was addressed by some users. Whereas this material poses an immense challenge, forcing us to rethink whether this material should be used for stable state or accept its ephemeral nature.

5.3 Material Physical Envelopes

The material composite plays a big role in this setup, even when considering a simple linear reading from the sensors. The composite becomes the encoding of the interaction modalities. This phenomenon could have been further investigated by working with other different covers. Therefore, the analysis between the freeform, elastic and plastic cover setups displayed how the material container does not only hold fluid; it acts as a physical filter for the haptic feedback.

Users were able to tell which gestures and how it affected the material to lead to a particular auditory effect. While metaphors played a role in how the

material affect the audio to translate possible interactions familiar to the users. This includes how it behaves to certain gestures, what users are perceiving when acting with the material and how it is associated with the resulting medium they are manipulating.

In its freeform state, the interaction is characterized directly through touch. The user experiences the entire sensorial spectrum of the material, including its binding qualities and stickiness. However, this high sensorial feedback quality comes at the expense of predictability. This is seen in descriptors such as “glitchy” or “blips”; expressed by P1 and P5.

On the other hand, when introducing the elastic and plastic covers, the experience shifts towards a mediated material interaction; moving away from a direct contact with the sample. The covers filter the material’s stickiness while putting emphasis on its resistance and phase-changing properties. There are situations when it introduces new qualities, such as the force-back, where it works in tandem with the internal non-linear resistance of the Oobleck.

Thus, this creates a hybrid profile of its haptic and digital feedback. The surface layer provides predictable elasticity, while the core inherently contains non-Newtonian resistance that hardens under kinetic forces. Therefore, the containments should not be seen as a protective layer that protects the prototype from its messiness. But rather it must be viewed as a physical filter capable of sculpting, smoothing or amplifying affordances to align with the targeted composite.

6 Future Work

This material presents lots of future possibilities of implementations within the domain of audio production. However, there are other areas that should be pursued as well in the future.

Although most of my thesis contributed on how the inherent feedback of the material reinforces user actions; resulting in a complete feedback loop between the audio and the user. This can go beyond just sound, but also how you approach any type of tangible interaction with Oobleck.

6.1 Interactivity & Possible Mediums of Application

In the domain of interaction design, I consider that Oobleck should be further researched in relation to visual mediums as well as through other medium based interfaces; e.g. olfactorily, haptic. This future research area could be valuable for other interaction designers to use this tangible abstract fluid to represent other mediums.

For example, the elastic composite dual interactive properties could change between temperature of a light and brightness by tapping, while presses in the inner composite could change the parameters. However, the inherent affordances become apparent separating user's involvement in two feedback qualities, where I consider as valuable as the entire range of possibilities of states and behaviours can be created by involving different surfaces.

Therefore, it is important to know how the medium can be represented more through this non-linear material. For example, if you press a button or a switch knob to trigger a change, usually they would have mechanical feedback; an on/off quality. Whereas Oobleck depending on the force could be both on/off feedback and rounded change by e.g. having an elastic surface to give linear feedback between min and max. Thus, there is necessity to designing digital constraints between afforded gestures, to limit dissociations between resulting qualities and sound.

6.2 Phase-Changing Interactive Materials

In terms materials, future research could also focus on mitigating temporal decay and its inconsistencies caused by evaporation and precipitation. This would need experts in the field of material science. Therefore, synthesizing a non-Newtonian fluid that retains the qualities but also change the transition speed of those states.

This became a future consideration in exploring other similar behaving materials or materials that are vastly different but non-linear in nature. E.g. a speculative material that could shift inversely to how Oobleck behave, or something that changes qualities such as thickness of the liquid from the speed of the movements done in the liquid.

Arguably I see the use of the metaphors as a translation of specific interactions associated to a medium that is being controlled. This is visible on how I used for example the force-back and the slow pacing of the composites as translation of previous practices within a context of use; e.g. modulating continuously an effect, holding in an effect.

6.3 Sensors & Physical Affordances

For digital hardware and materials, relying only a single input FSR sensor at the bottom limits the capacity to read complex interaction on a bigger scale. This could go further on larger surfaces holding multiple sensors, where you can change multiple ranges of qualities on different points; like a touch screen with dynamic inherent feedback derived from the Oobleck.

However, one aspect that could also direct could be the change between a linear to a non-linear mapping. A future direction should shift into machine learning mapping models of each interaction modality. This could adapt dynamically to the fluid's viscosity, while also smoothing and preserving multitude of expressions from one input.

This can also range from pulling, stretching the material, manipulating the Oobleck by introducing other elements, e.g. water, to control its state. Therefore, not only on downward pressures, but also previously mentioned forms of manipulating the material. Of course, this depends on the designer's decision on what surfaces are appropriate for the composite to afford such physical manipulations.

6.4 Future Music Producing Applications

A possible application for this material could be a digital musical instrument to generate sounds, not only to modify already produced content. However, considering the delay between attacks, user's need appropriate readjustments between different sounds. For example, using a drum kick generates a short, triggered audio after the attack following Oobleck's solid behaviour, but a flute or another longer transient attack would follow other more liquid behaviours.

Consequently, it is necessary to make an entire library for each existent sound to follow behaviour and states the material offers, which currently is not feasible to work with. Thus, future researchers, music instrument makers and audio interface designer could continue to expand this area of research.

However, I am not deterministic on what and how things could be approached. But there are some gaps in this current stage of my research that questions whether this could expand further in other audio production practices, e.g. Foley, synchronizing and manipulating audio samples. Therefore, I leave you, the reader, to conclude and speculate on other possible applications in this practice.

7 Conclusion

Considering the overall body of work, this thesis has explored the non-linear, sheer-thickening properties of Oobleck. Thus, expanding it as a TUI for post-production audio workflows and addressing its possibilities. Through the implementation of a combined methodological framework using Research through Design, Material Driven Design and Interaction Driven design.

Throughout the design process and its design cycles, the material composites displayed rich interaction modalities from music producers and students. The material successfully bridges the gap between user's motor actions and the auditory outcomes by producing a continuous feedback loop.

When users applied sudden force, the fluid hardens resulting on a kinetic force that dramatically output sonic transformations, such as abrupt panning

changes. Conversely, gentle, sustained movements afforded the material to sustain its fluidity, thus facilitating continuous modulations.

However, the direct interactions of the Oobleck at its free from also revealed design challenge. Primarily it manifested unpredictability and rapid temporal decay due to it precipitating and drying over time. Also, the first cycle deviated the process to centre for post-production practices, where sound is only manipulated with already existent audio content.

Conclusively, this work displayed both non-Newtonian fluids successfully support expression, but also failures due to its temporal decay. Thus, Oobleck supports embodied audio manipulations through its resistance profiles that linear material cannot replicate. Therefore, it opens for multiplicity of possibilities of highly expressive, improvisational and exploratory sound design practices.

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Appendix

Backup Drive (In case the GitHub links will broke eventually)

<https://drive.google.com/drive/folders/1FErAonWPz5lgQ-D2TQk2MuVQhDXxzGJ6?usp=sharing>

GITHUB PAGE with RAW SENSOR DATA and SKETCHES (If you want to out): <https://github.com/DrakeMech/ThesisAppendixContent>

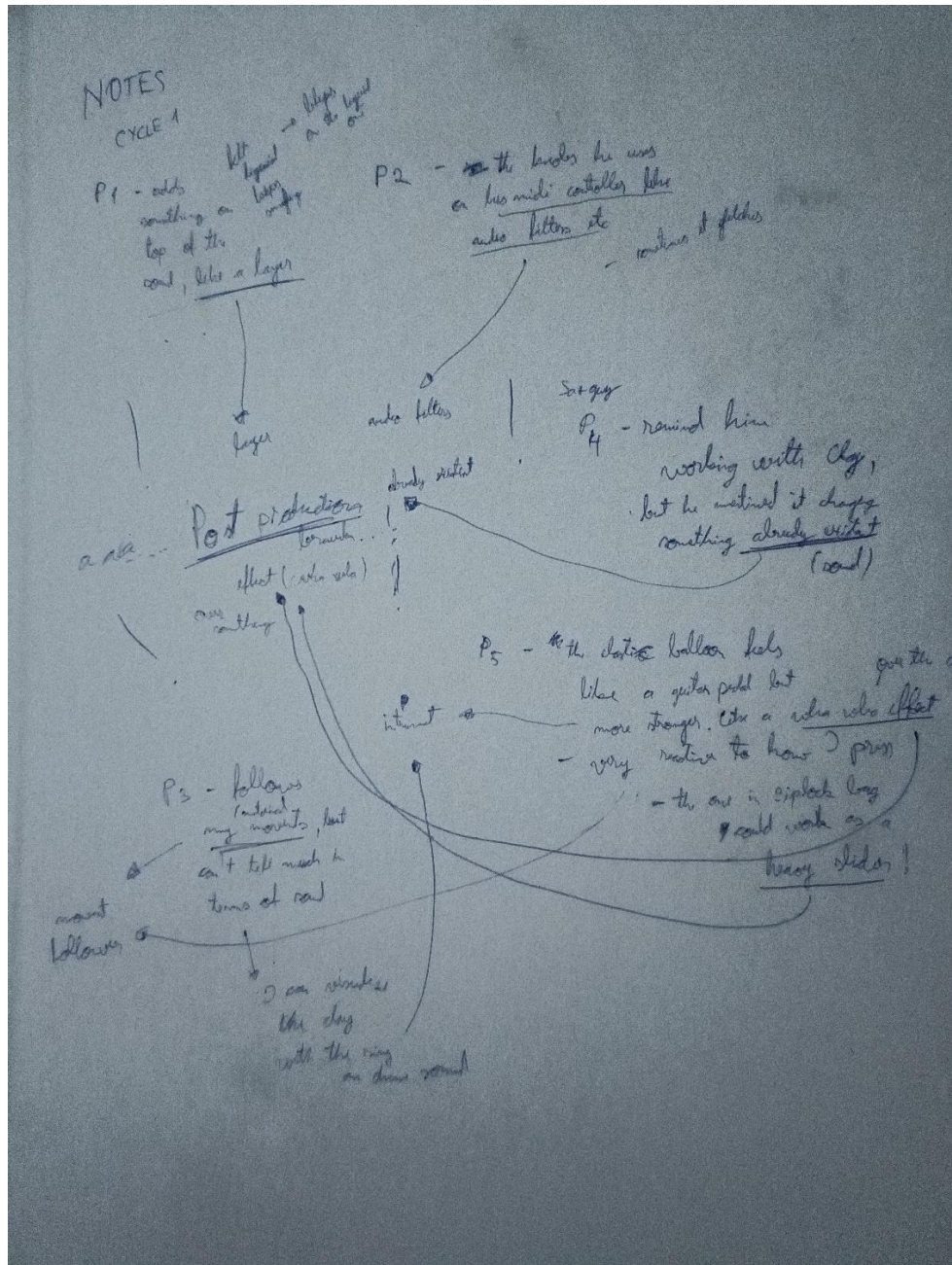
CONSENT FORM (It is in the drive as well)



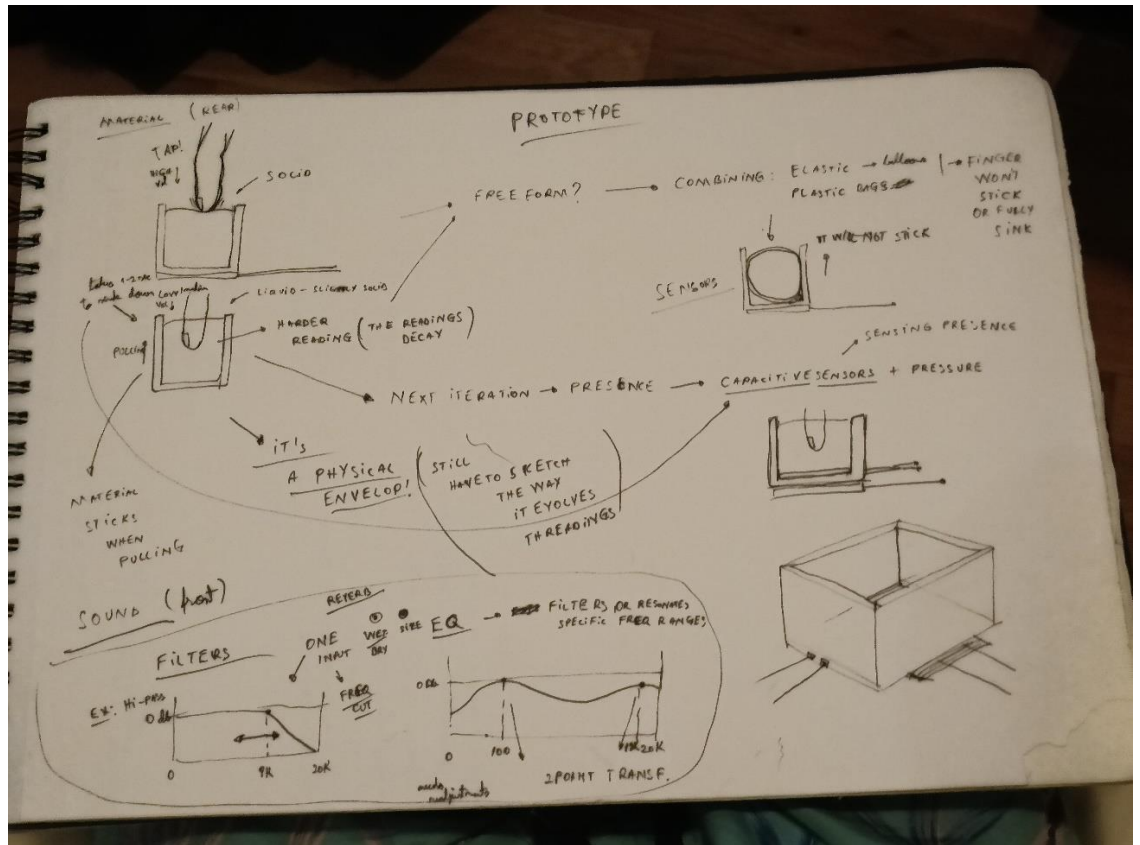
Information letter	Appendix 1
Project title: Non-Newtonian Sound Interfaces: Dynamic affordances of Oobleck as means of sonic manipulations.	Date: 15 Feb – 15 June 2026
Study manager: David Aleklett Kadish david.kadish@mau.se Your e-mail Stefan Cristache Limit_overthe@yahoo.com	Studying at Malmö University, Faculty of Culture and Society, S-205 06 Malmö, Phone +46 40 665 70 00 Education: Interaction Design Level: Bachelor of Science
Who am I? My name is Stefan Cristache. I am a student of Interaction design at Malmö University. I am currently working on my bachelor's thesis under the supervision of David Aleklett Kadish. This is for processing the documentation gathered as result of the test that is going to be conducted for you, the participant.	
What are my objectives? My aim is to demonstrate how the dual dynamic affordances of Oobleck support audio manipulation in audio production. This will take a further take into the situations, specifically during postproduction, to gather data on how these interactive dynamics play between you and the material.	
What is Oobleck? Oobleck is a mixture of cornstarch and water. It is a non-Newtonian fluid, a fluid that is a shear-thickening liquid that changes states between a solid and a liquid based on the physical stress absorbed by it.	
Why is that important? The Oobleck will be used as a physical interface for audio manipulation. Considering that aspect, it is essential for you, the participant, not to have any allergies to starches. Additionally, since it's a viscous liquid, you will get directly in contact with the Oobleck, which can make some participant uncomfortable.	

FIRST CYCLE

NOTES (You can also check in the drive for better view)

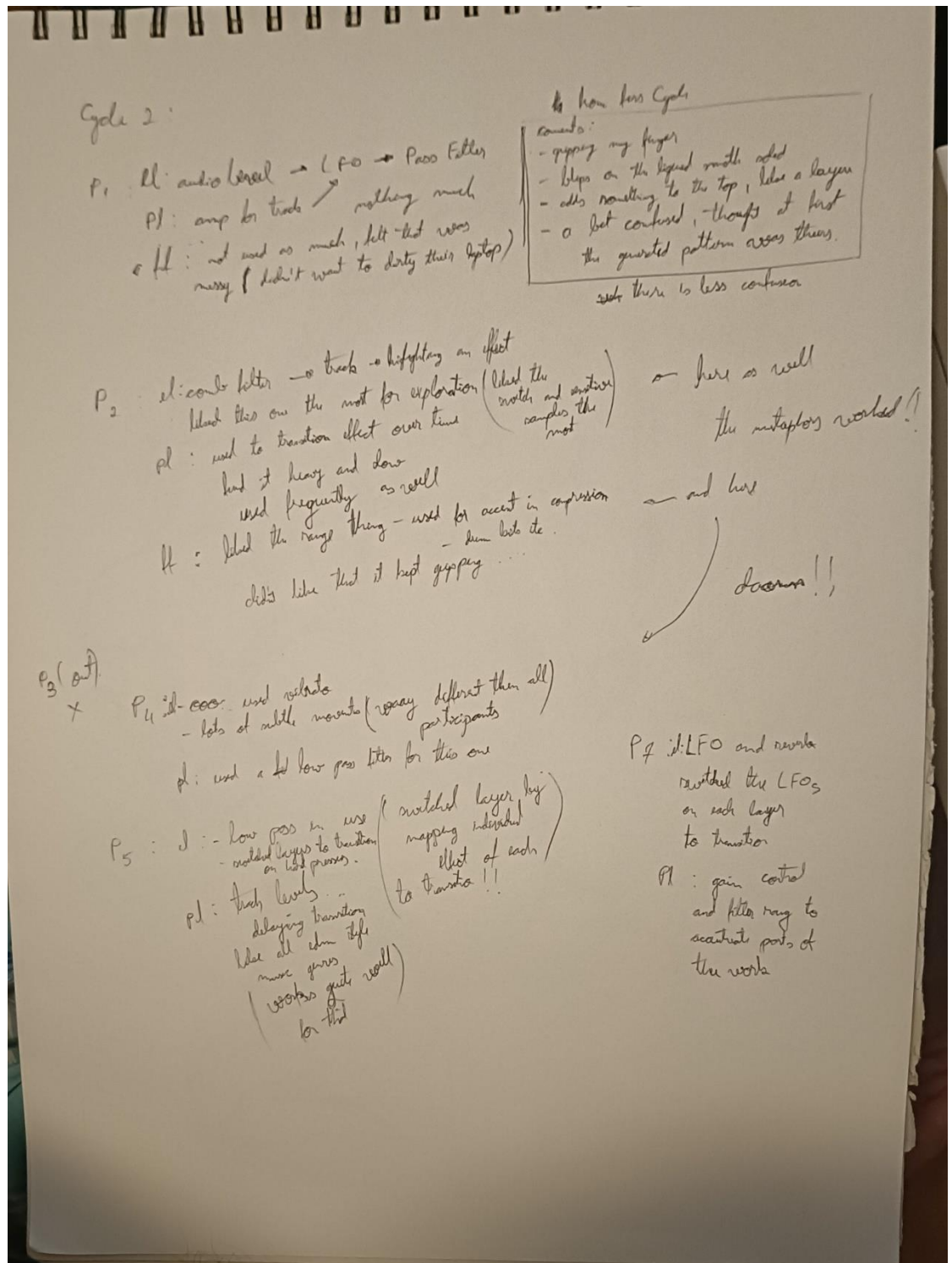


SKETCHES (You can also check in the drive for better view)



SECOND CYLCE

NOTES (You can also check in the drive for better view)



SKETCHES for the Metaphors

